AI Travel Planner Business Use Case

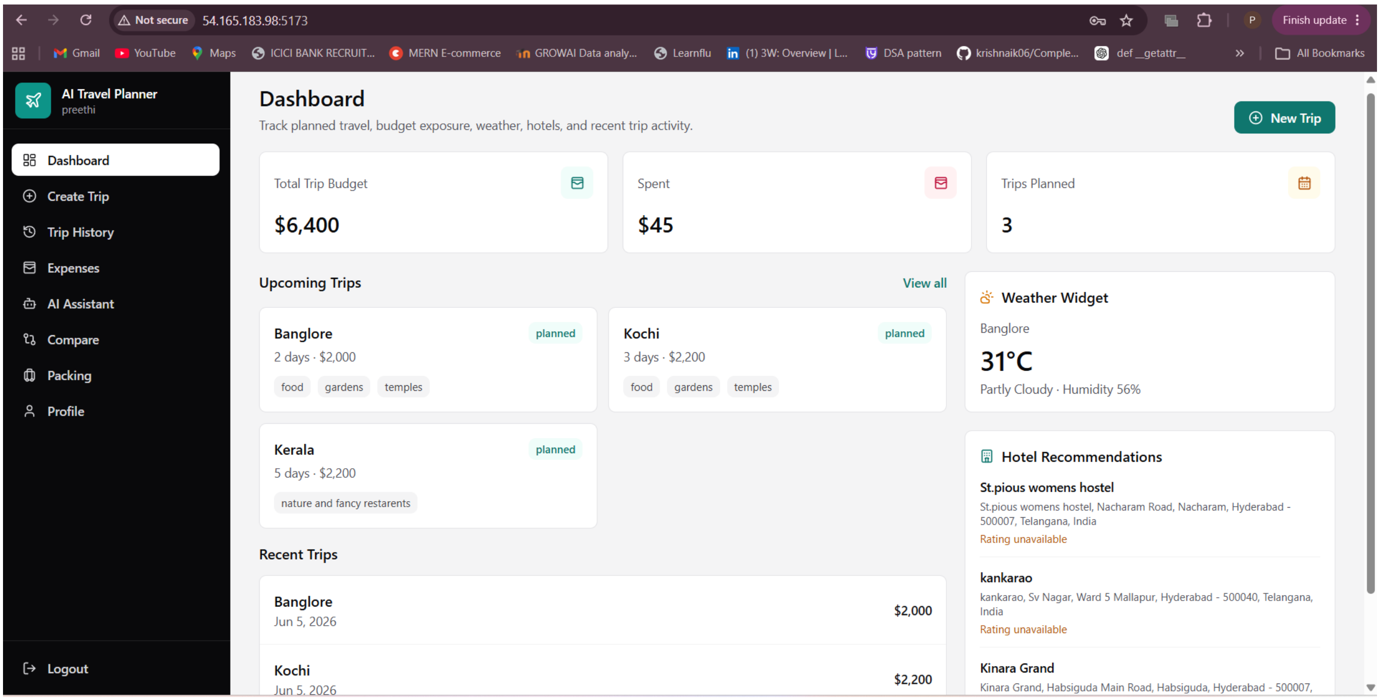
Business Requirement

AI Travel Planner is a cloud-native travel management platform that helps users plan trips, generate AI-powered itineraries, manage travel expenses, compare destinations, and receive travel recommendations.

The application leverages Artificial Intelligence to create personalized travel plans based on destination, budget, trip duration, and user interests. It also assists travelers with destination comparison, expense tracking, packing recommendations, and travel guidance.

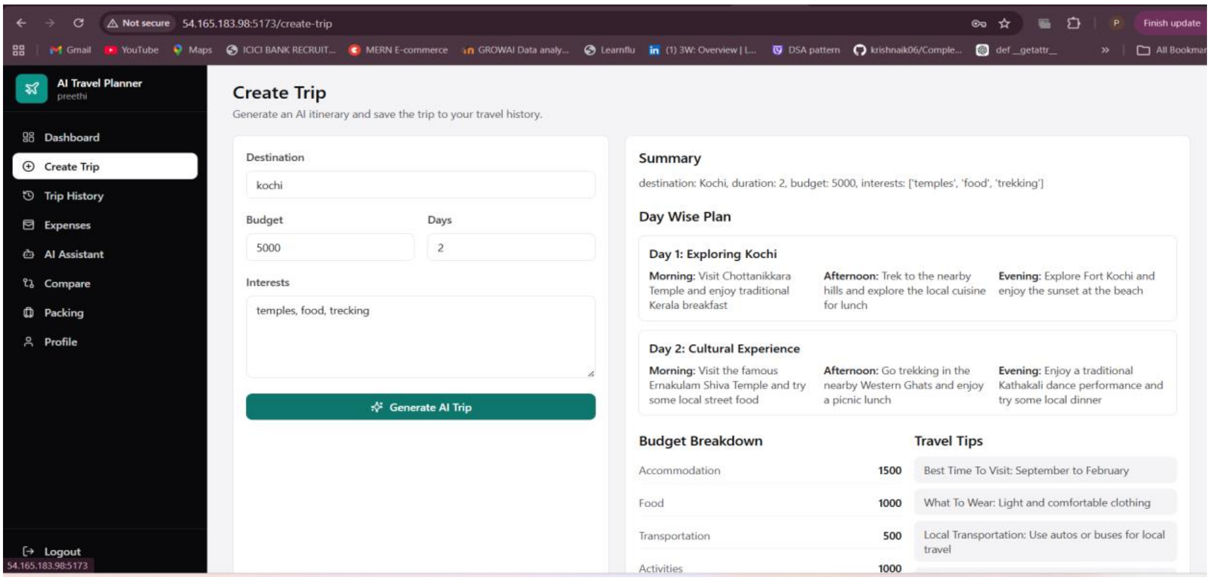
Primary Business Value:

• Generate AI-powered travel itineraries
• Track and manage travel expenses
• Compare destinations using weather, activities, and cost data
• Provide intelligent travel assistance through an AI chatbot
• Generate destination-specific packing checklists
• Store travel history and planned trips
• Secure application secrets using AWS Secrets Manager
• Encrypt sensitive application data using AWS KMS
• Archive application logs into Amazon S3 for auditing and troubleshooting
• Deploy a scalable cloud-native application on AWS EC2

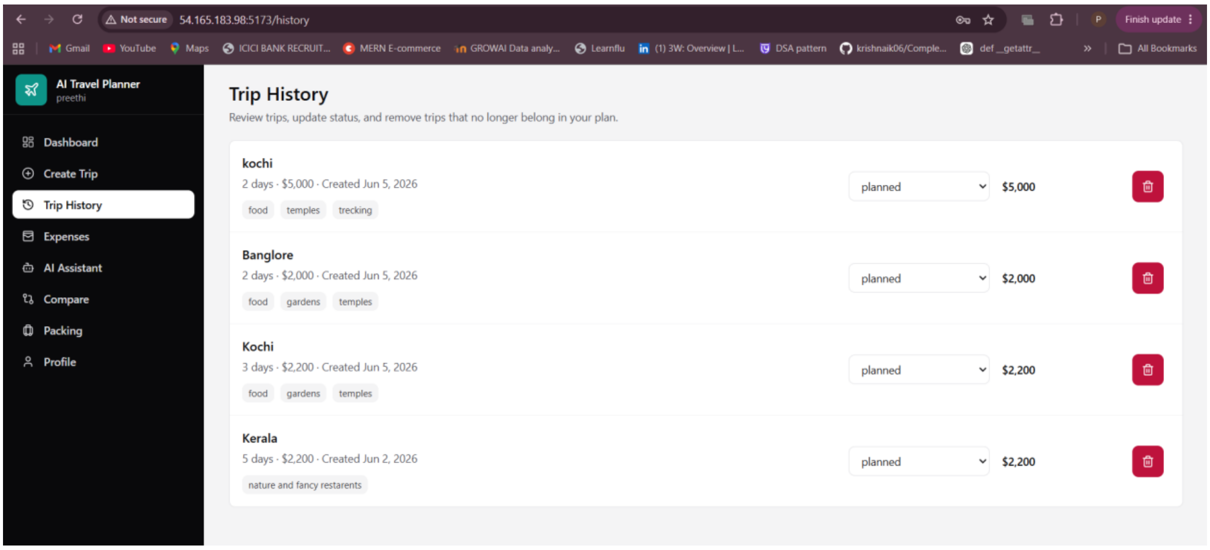


Application Features

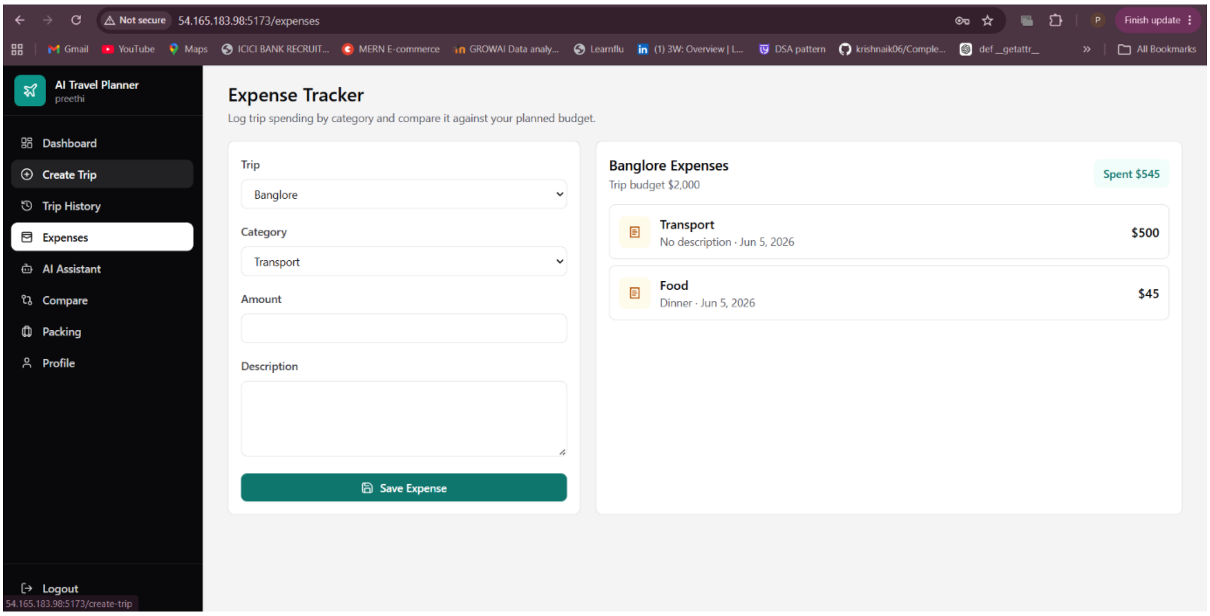
1. AI Trip Generation
Users can generate personalized travel itineraries based on destination, budget, duration, and interests. The AI engine produces day-wise travel plans, budget breakdowns, and travel recommendations



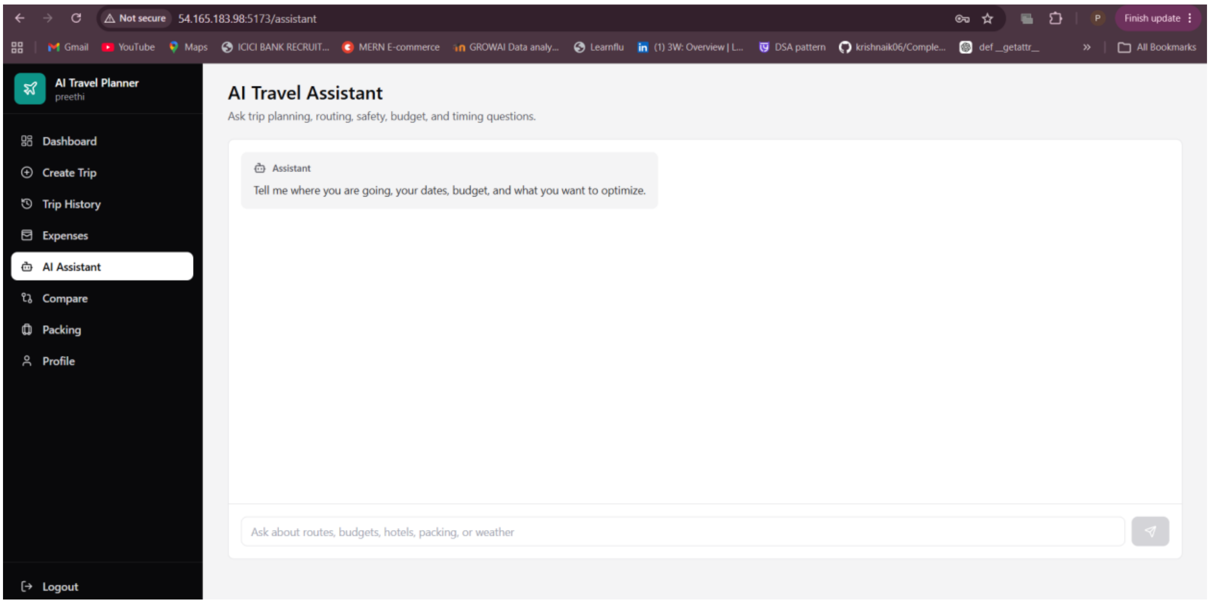
2. Trip History Management
Users can review previously planned trips, update trip status, and manage travel records.



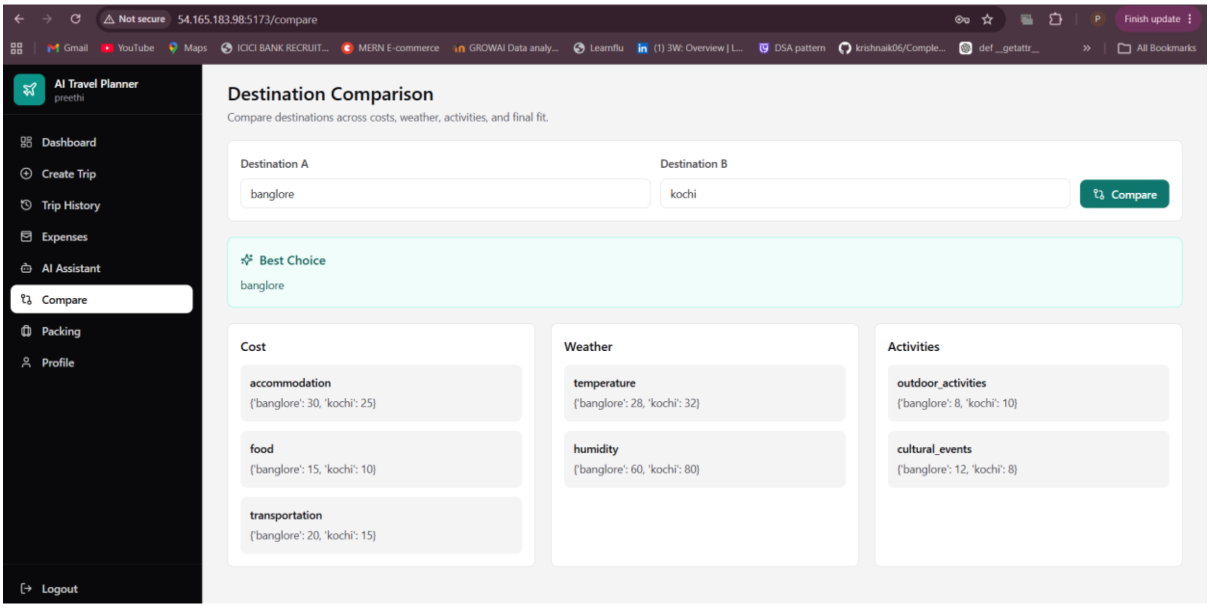
3. Expense Tracking
   The application allows travelers to record expenses and compare actual spending against planned budgets.



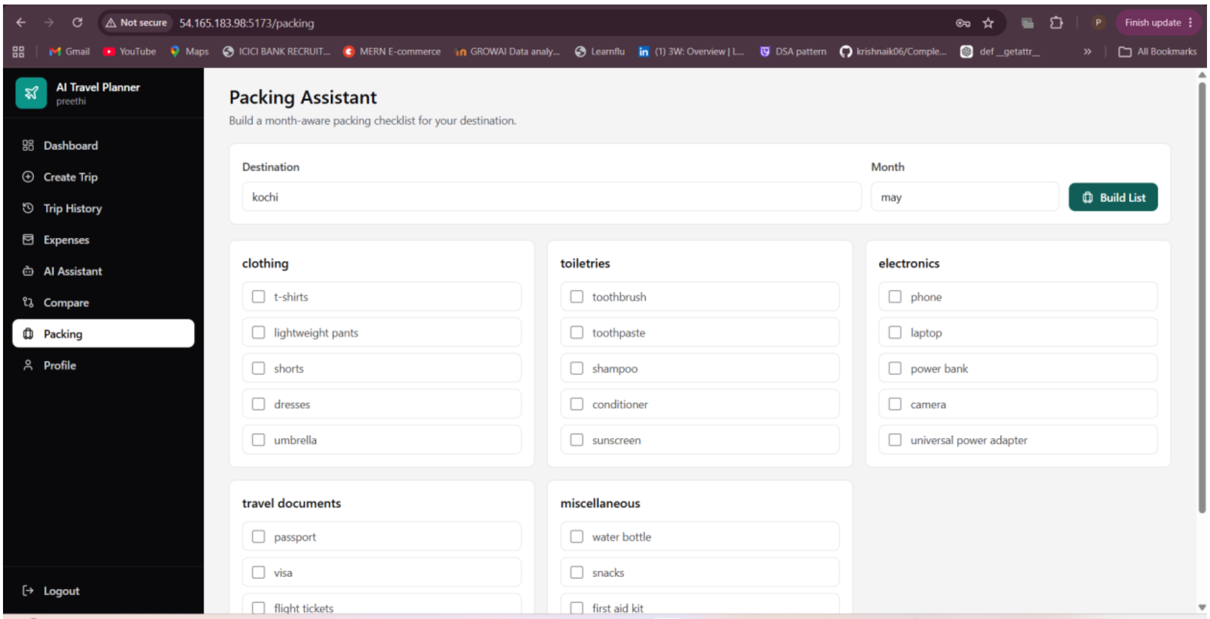
4. AI Travel Assistant
   Integrated AI assistant helps users with travel planning, budgeting, routing, weather information, and destination recommendations.



5. Destination Comparison
   Users can compare destinations using weather conditions, activity availability, and travel costs to identify the most suitable destination.



6. Packing Assistant
   Generates customized packing checklists based on destination and travel month.



Deployment Architecture

The AI Travel Planner application is deployed on an Amazon EC2 instance.

Application Components:

- Frontend – React
- Backend – FastAPI
- Database – PostgreSQL
- Secrets Management – AWS Secrets Manager
- Encryption – AWS KMS
- Log Storage – Amazon S3
- Hosting Platform – Amazon EC2

The application was successfully deployed and tested end-to-end, including trip generation, expense tracking, destination comparison, AI assistant functionality, secret retrieval, KMS integration, and S3 log archival.

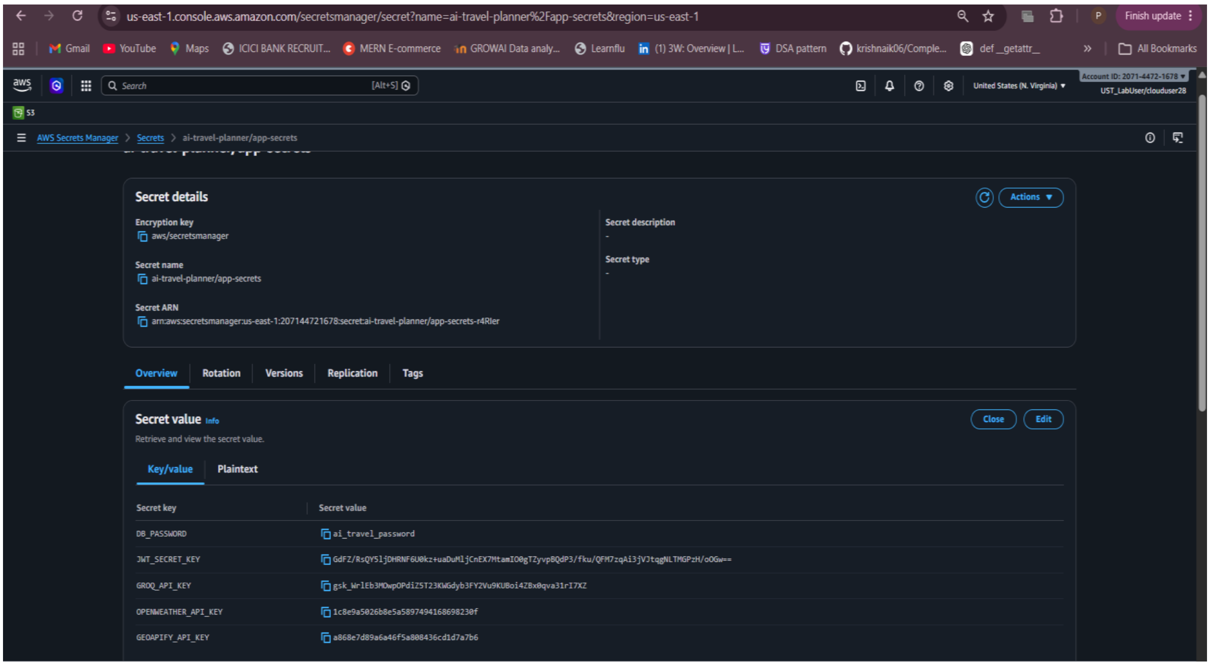
AWS Services Used

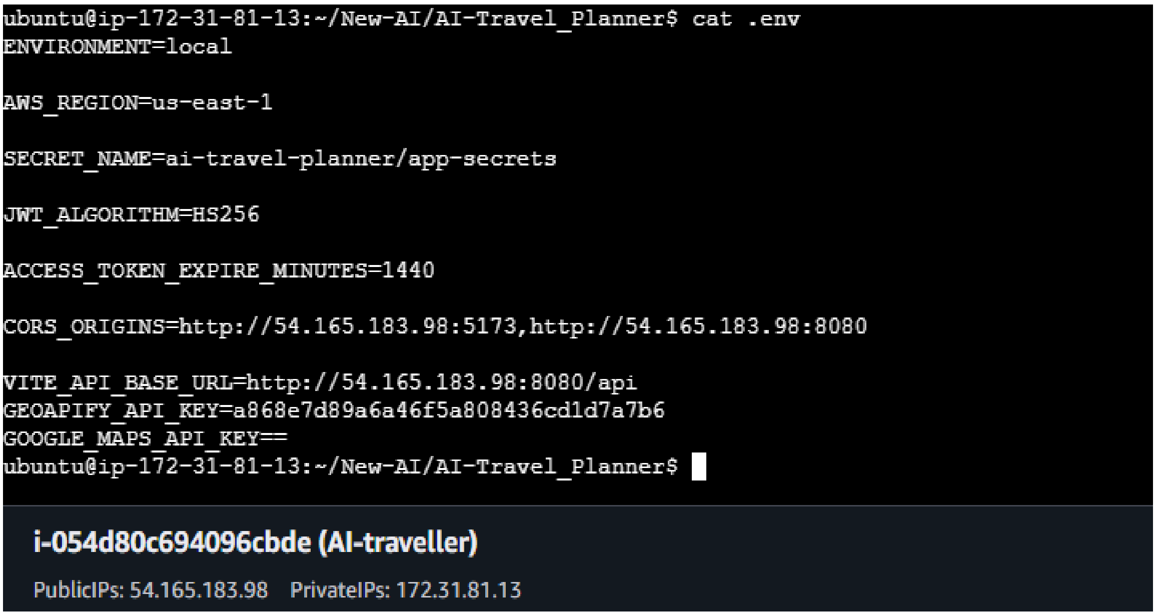
PostgreSQL

PostgreSQL is used as the primary database for storing user accounts, trip history, expense records, destination comparisons, and itinerary information.

The database provides structured storage and allows users to retrieve travel plans even after application restarts.

AWS Secrets Manager





Implemented AWS Secrets Manager to securely store and retrieve application secrets and configuration values.

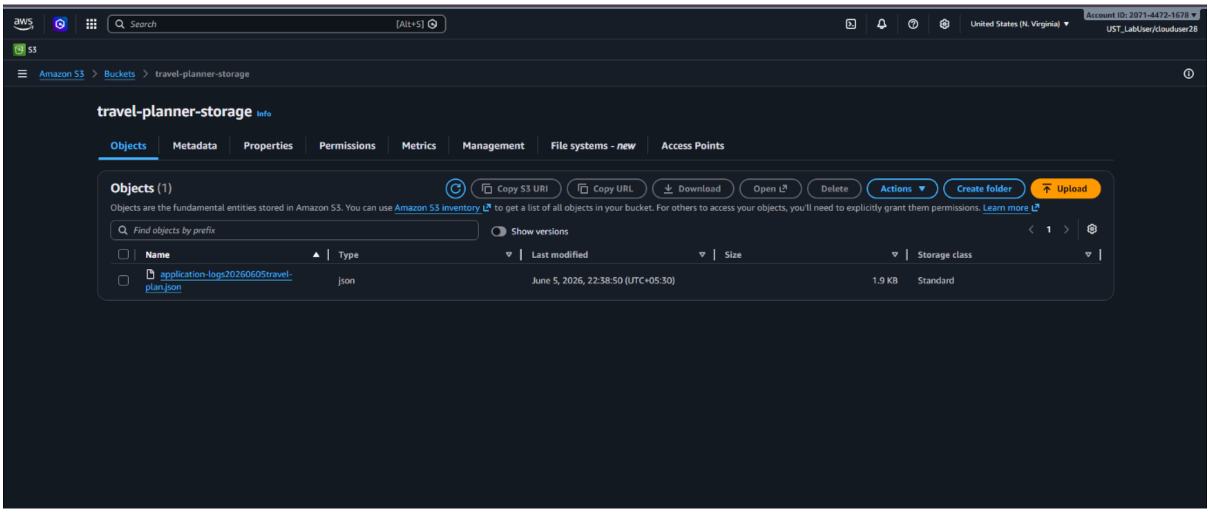
The following sensitive information is stored securely in AWS Secrets Manager:

- Database Password
- JWT Secret Key
- GROQ API Key
- OpenWeather API Key
- Geoapify API Key

The application retrieves secrets dynamically during runtime using AWS SDK integration. No sensitive credentials are hardcoded in the source code.

This approach improves application security and simplifies secret management across environments.

Amazon S3 Log Storage




```
travel-planner-storage
application-logs20260605travel-plan.json
```

Implemented Amazon S3 as a centralized storage location for application logs.

The application periodically archives operational logs and activity records into an S3 bucket named travel-planner-storage.

Benefits of S3 log archival include:

• Long-term log retention

• Centralized log management

• Audit and compliance support

• Cost-effective storage

• Disaster recovery support

Log files are stored as JSON objects within the S3 bucket, enabling future analysis and troubleshooting of application events.